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Experimental investigation of water spray injection in liquid piston for near-isothermal compression

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HIGHLIGHTS

- Experimental study using water spray injection in a liquid piston air compressor.
- The effect of injection pressure, nozzle angle, and compression speed is studied.
- Isothermal compression efficiency increases up to 95% with an increase of injection pressure.
- Nozzle angle and compression speed have a marginal effect on the isothermal efficiency.
- Spray injection is a highly effective method to achieve a near-isothermal compression.

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ABSTRACT

Near-isothermal compression is desired to achieve high efficiency in many compressor applications. Low heat transfer characteristic of conventional compressors is a major bottleneck in attaining a near-isothermal compression. A high heat transfer rate is possible with an injection of a large number of water droplets using a spray nozzle inside the compression chamber. In this paper, the effectiveness of spray injection to achieve near-isothermal compression is investigated experimentally in a liquid piston compressor for a compression ratio of about 2.5. Parametric investigations are performed by varying injection pressures of spray from 10 psi (69 kPa) to 70 psi (483 kPa), using different spray nozzle angles (60°, 90°, and 120°), and by changing the stroke time of compression. It is observed that water spray injection is highly effective in abating the air temperature rise during the compression process. The pressure-volume plots indicate a significant reduction in the compression work, and they approach near-isothermal compression with spray at higher injection pressures. The isothermal efficiency of compression consistently increases with an increased injection pressure of spray and reaches up to 95% at the highest injection pressure studied (70 psi). Furthermore, the spray nozzle angle marginally affected the isothermal efficiency with a 1–4% improvement with the use of a 60° nozzle angle over a 120° spray angle at all injection pressures. Also, comparable isothermal efficiencies are observed for compression with different stroke times between 3 and 5 s especially at higher injection pressures which highlight the efficacy of spray injection in attaining a high power-density along with high efficiency. Overall, with an optimized spray design, water spray injection can achieve a highly efficient near-isothermal compression in liquid piston.

1. Introduction

Renewable energy resources play a vital role in reducing energy dependence from fossil fuels. These resources fulfill vital energy needs in the face of rising energy demands across the world and the devastating effects of global warming. However, due to the nature of renewable energy sources such as solar, wind, wave, and tidal energy; significant inherent fluctuations make it harder to effectively integrate them into the electric grid. Large-scale energy storage systems are needed to facilitate effective utilization of renewable energy in the electric grid [1]. Various electrical energy storage methods such as high-power density batteries, fuel cells, hydroelectric storage, and compressed air energy storage (CAES) systems have been explored. Apart from hydroelectric power storage, energy storage is not widely prevalent on a utility grid-level [2]. CAES systems can provide a viable alternative to serve as large-scale energy storage systems. Efficient energy storage systems facilitate the effective utilization of intermittent renewable energy sources with minimum energy losses. To achieve a

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Nomenclature

A	area (m^2)
C	a constant
A_d	surface area of a droplet (m^2)
A_d	surface area of a droplet (m^2)
$A_{d,i}$	surface area of i^{th} droplet (m^2)
$A_{d,o}$	overall surface area of all the droplets (m^2)
C_v	specific heat of air at constant volume (J/kg K)
h_d	heat transfer coefficient between a droplet and gas (W/m^2 K)
$h_{d,i}$	heat transfer coefficient between i^{th} droplet and gas (W/m^2 K)
k	heat capacity ratio of gas
m	mass of the gas (kg)
n	polytropic index of compression process
N	number of droplets
P	pressure (Pa)
P_0	pressure at the start of compression (Pa)
P_f	pressure at the end of compression (Pa)
P_r	compression pressure ratio
$\dot{Q}$	rate of heat transfer (J/s)
$\dot{q}_d$	rate of heat transfer to a single droplet (J/s)
$\dot{Q}_d$	rate of heat transfer to N droplets (J/s)
R	gas Constant (J/kg K)
t	time (s)
T	temperature of gas (K)
T_∞	temperature of ambient (K)
T_d	surface temperature of a droplet (K)
$T_{d,i}$	surface temperature of i^{th} droplet (K)
T_g	temperature of gas (K)
$\dot{U}$	rate of change of internal energy of gas (J/s)
U_h	overall heat transfer coefficient (W/m^2 K)
V	volume (m^3)
V_0	volume at the start of compression (m^3)
V_f	volume at the end of compression (m^3)
V_{iso}	volume at the end of compression with isothermal process (m^3)
$\dot{W}_{comp}$	compression work (J/s)

Greek symbols

η_{iso}	isothermal efficiency of compression

Subscripts

0	at the start of compression
d	droplet
g	gas
i	identification number of the droplet
r	ratio
iso	isothermal
comp	compression

commercially viable energy storage system, CAES systems need to be highly efficient along with providing a high power density at the same time [3].

In a typical CAES plant, the air is used as an energy storage medium by compressing it using an air compressor and then expanding it at peak time over a gas turbine to generate electricity. Isothermal CAES is shown to be highly efficient without the need for fuel and thermal energy storage [4]. The compressor and expander are key components in an isothermal CAES and they should operate isothermally to obtain a high roundtrip efficiency [5]. Many novel energy storage systems like ocean compressed air energy storage (OCAES) [6], hydro-pneumatic compressed air ground-level integrated diverse energy storage (GLIDES) [7], near-isothermal-isobaric compressed gas energy storage [8], and open accumulator isothermal compressed air energy storage (OA-ICAES) [9] use isothermal compressor and expander in the design of an efficient energy storage system. Typically for an isothermal compression/expansion, the process involves a gradual pressure change while allowing continuous heat transfer to maintain a constant temperature during the process. Since in any practical industrial application it is difficult to achieve an isothermal compression and expansion, innovative techniques to attain a high heat transfer rate are needed in the compressor/expander to approach a near-isothermal process. Overall, the development of a near-isothermal compressor and expander would make CAES systems highly efficient, economical, and competent energy storage [10].

Typical air compressors for industrial applications tend towards an adiabatic process with a high air temperature at the end of compression. This is because a significant proportion of the compression work is utilized in increasing the internal energy of the gas. Reducing the temperature rise of gas during the compression process decreases the compression work and improves compression efficiency [11]. Liquid piston gas compression has the potential to obtain significant improvements in compression efficiency compared to conventional reciprocating compressors [12]. In a liquid piston compressor, a column of liquid acts as a moving piston to compress the gas to required pressures in a given chamber volume. Improving the heat transfer characteristics in a liquid piston plays an important role in increasing the efficiency of the liquid piston compressor [13]. It is observed that the heat transfer coefficient in a liquid piston compressor is high at the initial phase of compression but decreases towards the later phase of compression. Additionally, the heat transfer coefficient increases marginally with an increase in the piston velocity [14]. To achieve an isothermal compression, the instantaneous compression work should be equal to the instantaneous heat transfer rate. This demands a very high heat transfer rate from the gas to the liquid and its surroundings in the liquid piston.

Heat transfer analysis of liquid piston compression for hydrogen applications was studied by Kermani et al. [15]. Their sensitivity analysis suggested that the heat transfer coefficients and compression time were important factors in reducing the hydrogen temperature. Heat transfer enhancement in liquid piston compression can also be achieved by optimizing the trajectories for the compressor. A 10–40% increase in power density was observed by optimizing the piston profile while also accounting for the effect of viscous friction [13]. Zhang et al. studied the heat transfer during compression with porous media inserts in a compression chamber [16]. The porous medium allows the gas and liquid to pass through while increasing the net surface area available for heat transfer, and thus reducing the temperature rise during the compression process. Improving the heat transfer through porous media inserts improved the efficiency of the compression process which shifts the polytropic process towards a near isothermal process. Yan et al. conducted an experimental study on the heat transfer enhancement in a liquid piston using porous media inserts to quantify its effects in terms of power density [17]. They observed that with the use of porous media inserts, the power density increased by 39 times at 95% efficiency and the efficiency increased by 18% at 100 kW/m^3 for the compression process. A similar study at high pressure showed a tenfold increase in power density at a constant efficiency during the compression and expansion processes [18]. The major contributing factor for increasing the heat transfer from the liquid piston compressor/expander was the increase in surface area available due to the porous media inserts. Moreover, a novel heat transfer enhancement using aqueous foam in a liquid piston compressor showed up to 92% compression efficiency [19]. A large surface area and a high heat transfer coefficient of aqueous foam were useful in achieving a high heat transfer rate in the compression chamber.

A varied approach to improve the heat transfer or reduce the temperature rise of the compressing gas is to inject water droplets during the compression process. For a liquid piston compressor with water as the compressing liquid, water spray injection is miscible while at the same time it uses the high specific heat of the water to absorb the heat and reduce the temperature of the gas. One dimensional simulation of a liquid piston compression with droplet heat transfer indicates that compression efficiency can be increased from 71% for adiabatic compression up to 98% with spray injection for a tenfold compression ratio [20]. Several researchers have studied droplet dynamics and heat transfer between air and water spray. Experiment and numerical studies on water spray and ambient air to study evaporative cooling are performed by Sureshkumar et al. [21,22]. They investigated spray cooling for parallel flow and counter-flow configurations with different nozzle pressures, nozzle diameters, environmental conditions, and air velocities. Accordingly, they observed that, for a given flow rate, higher injection pressure with a smaller nozzle diameter resulted in a higher temperature drop of air. Since the goal of water spray injection is to minimize the temperature rise, the amount of water injected, and the droplet dynamics are significantly important. Coney et al. described a reciprocating piston compressor using water spray injection to achieve a near-isothermal air compression [23]. They observed a substantial reduction in the air temperature and the compression work by using high mass loading of water. The injection of spray during compression results in a significant increase in heat transfer out of the system and a quantifiable drop in the temperature of the air during the compression process. Therefore, injection of water spray in a liquid piston compressor shows great potential to achieve a near-isothermal compression. Furthermore, it is convenient to accommodate water spray injection in a liquid piston compressor as the water droplets eventually fall in the liquid piston surface and no other mechanism is required for removal of extra water from the chamber compared to a convenient reciprocating compressor. The investigation of spray injection in liquid piston is so far limited to the computational studies and to the best of our knowledge, water spray injection has not been studied experimentally thoroughly in a liquid piston compressor setup.

In this paper, water spray injection is studied experimentally in a liquid piston compressor to investigate its effectiveness to achieve a near-isothermal compression. The injection pressure of spray, nozzle angle of spray, and compression stroke time are varied to examine the effect of these parameters on the compressor performance. The conceptual framework of isothermal compression and spray injection is presented in the next section. Then, details about the experimental setup of the liquid piston with spray injection and a range of experiments performed are presented. Furthermore, the effects of injection pressure of spray, nozzle angle, and stroke time of compression on temperature, pressure-volume plot, and isothermal efficiency of compression are discussed. Finally, the overall conclusions and possible future extensions of this study are stated.

2. Conceptual framework of isothermal compression and spray injection

The liquid piston compression process involves a moving column of liquid that compresses a finite gas volume in a compression chamber. During the compression process, the liquid enters the compression chamber and reduces the available gas volume, thereby, increasing the gas pressure and temperature. The control volume of gas in the liquid piston compressor is shown in Fig. 1. The compressed gas is sent out to the compressed air storage vessel through an outlet port. The inlet port then fills the chamber with the gas from the atmosphere to attain its initial conditions for the next cycle to start while the piston retracts to its original position.

Applying the first law of thermodynamics to the gas in the control volume (neglecting changes in kinetic and potential energies) can be expressed as:

$$\dot{U} = \dot{Q} - \dot{W}_{comp} \tag{1}$$

where $\dot{U}$ is the rate of change of internal energy, $\dot{Q}$ is the rate of heat transfer and $\dot{W}_{comp}$ is work of compression.

Using expressions for internal energy and compression work for a closed system in Eq. (1), we get

$$mC_v \frac{dT_g}{dt} = \dot{Q} - P\frac{dV}{dt} \tag{2}$$

where m is mass of the gas, C_v is the specific heat of the gas at constant volume, T_g is the average bulk temperature of the gas, t is time, P is the pressure of the gas, and V is the volume of the gas.

During the compression process, the term for compression work $\left(P\frac{dV}{dt}\right)$ is always negative in sign due to a decrease in the gas volume with an increase in time. For an isothermal compression, the rate of change of temperature should be zero which is the left term in Eq. (2). This requires the rate of heat transfer ($\dot{Q}$) to be equal to the compression work $\left(P\frac{dV}{dt}\right)$. In general, the rate of heat transfer in the liquid piston compressor is significantly less than compression work which results in a significant rise in the temperature of the air. Heat transfer enhancements are needed to achieve a near-isothermal compression process in a liquid piston compressor.

Thermal analysis of a liquid piston compressor [14] indicates that convective thermal resistance inside the chamber, the conductive thermal resistance of chamber material, and convective thermal resistance of surrounding outside the chamber contribute to the overall thermal resistance of a liquid piston compressor. It was observed that convective thermal resistance inside the chamber has a significant contribution to the overall thermal resistance compared to the other two resistances. Therefore, the heat transfer mechanism which reduces the convective thermal resistance between gas and chamber will be a major contributing factor in improvement of the overall heat transfer rate in the liquid piston compressor. Injection of water droplets in the compression chamber improves convection inside the chamber thereby

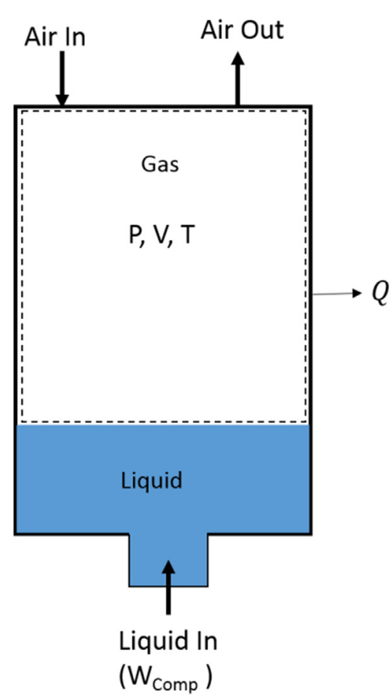

Fig. 1. Gas control volume in liquid piston compressor.

reducing the convective thermal resistance of gas significantly. Therefore, the water spray injection method has the potential to improve overall heat transfer in a liquid piston compressor.

The rate of heat transfer in a liquid piston compressor can be expressed as [14]

$$\dot{Q} = U_h A (T_\infty - T_g) \tag{3}$$

where U_h is the overall heat transfer coefficient, A is the area of heat transfer, and T_∞ is the temperature of the surrounding.

The rate of heat transfer from the gas to a single water droplet is given as

$$\dot{q_d} = \mathrm{h_d} A_d (T_\mathrm{d} - T_g) \tag{4}$$

where $\dot{q_d}$ is rate of heat transfer, $\mathrm{h_d}$ is heat transfer coefficient between droplet and gas, A_d is surface area of droplet, and T_d is average surface temperature of droplet.

With the injection of water spray, many droplets are introduced inside the compression chamber as shown in Fig. 2. The surface area of the droplets, velocity, and the temperature of the droplet vary from each droplet to droplet at any given instance of time. The overall rate of heat transfer from the gas to 'N' droplets is given as:

$$\dot{Q_d} = \sum_{i=1}^{N} \mathrm{h_{d,i}} A_{d,i} (T_{d,i} - T_\mathrm{g}) \tag{5}$$

where $\dot{Q_d}$ is overall rate of heat transfer from the gas to N droplets, $\mathrm{h_{d,i}}$ is heat transfer coefficient between $\mathrm{i^{th}}$ droplet and gas, $A_{d,i}$ is surface area of the $\mathrm{i^{th}}$ droplet, and $T_{d,i}$ is average surface temperature of the $\mathrm{i^{th}}$ droplet.

With the spay injection, the energy equation for the gas is given as

$$mC_v \frac{dT_g}{dt} = \left[U_h A (T_\infty - T_g) + \sum_{i=1}^{N} \mathrm{h_{d,i}} A_{d,i} (T_{d,i} - T_\mathrm{g}) \right] - P \frac{dV}{dt} \tag{6}$$

Assuming a constant heat transfer coefficient for all the droplets and a constant temperature of the droplets, the Eq. (6) can be simplified to

$$mC_v \frac{dT_g}{dt} = [U_h A (T_\infty - T_g) + \mathrm{h_d} A_{d,o} (T_d - T_\mathrm{g})] - P \frac{dV}{dt} \tag{7}$$

where $A_{d,o}$ is total surface area of all the droplets. The high number of droplets in the spray contribute to a high surface area resulting in an improved rate of heat transfer and which effectively helps achieve a near-isothermal compression.

The gas compression process can be assessed by considering the polytropic process of compression using the following relation.

$$\mathrm{PV^n = C} \tag{8}$$

where n is the polytropic index and C is a constant.

The polytropic index provides a reference for various types of processes and determines how close the process behaves compared to an isothermal process. The typical pressure-volume plots of compression are shown in Fig. 3. When there is no heat transfer from the gas to the surrounding (adiabatic process), the polytropic index equals to heat capacity ratio (k) of the gas. Improving the heat transferred from the gas shifts the compression process from adiabatic conditions towards an isothermal condition. A polytropic index close to 1 represents a near-isothermal process. For CAES applications, achieving a high-pressure gas without the accompanying high temperature is the primary objective. Therefore, the isothermal efficiency of a compressor (η_{iso}) is defined as the ratio of work required for isothermal compression to the actual compression work including the cooling work by the following equation [14].

$$\eta_{iso} = \frac{\ln P_r - 1 + \frac{1}{P_r}}{\frac{P_r^{\left(\frac{n-1}{n}\right)} - 1}{n-1} + P_r^{-\frac{1}{n}} - 1 + (P_r - 1)\left(P_r^{-\frac{1}{n}} - \frac{1}{P_r}\right)} \tag{9}$$

where P_r is compression ratio and the polytropic index n is $1 < n < \mathrm{k}$.

3. Experimental setup

Water spray injection is investigated experimentally for compression of air in a liquid (water) piston compressor. The effectiveness of spray to abate the air temperature rise is assessed from the experimental measurements of pressure, volume, and the temperature of the air in the compression chamber. Detailed descriptions of the liquid piston setup, spray injection setup, measurement accuracies for various measurement devices, and the range of experiments performed are discussed in this section.

3.1. Liquid piston setup

The experimental setup of the liquid piston compressor used for the experiments is shown in Fig. 4. The setup consists of a compression chamber, solenoid valves, intake and exhaust valves, a pneumatic cylinder, a hydraulic pump, controller, Data Acquisition System (DAQ), and measurement instruments. The outer cylinder contains water acting as a surrounding environment for the compression chamber. This helps maintain a constant surrounding temperature and acts as a larger heat sink during the compression process compared to air as the surrounding medium. Water from a hydraulic pump drives the liquid piston interface in the compression chamber which compresses the air and is retracted by a coupled pneumatic piston-hydraulic pump for the intake stroke. A cycle consists of a compression stroke followed by the exhaust of high-pressure air and ends with an intake stroke in which atmospheric air fills the chamber to reach the initial conditions. During the compression process, the liquid column moves in the upward direction and continuously reduces the air volume while increasing the pressure and temperature. Each experimental test case is run for about 10–12 continuous cycles to confirm repeatability and evaluate any cyclic variability. The controller actuates the solenoid valves which regulate the intake and exhaust of air at the end of compression. The exhaust valve is opened to the atmosphere when the piston in the pneumatic cylinder reaches the outlet end. This ensured a constant volume ratio of compression by adding a fixed amount of water in chamber at all the cycles. A LabVIEW [24] program was used for the continuous operation of the liquid piston. A polypropylene container was used as the compression chamber with a diameter of 11 cm and the height of the initial

Fig. 2. A representative model of spray-droplets in a liquid piston.

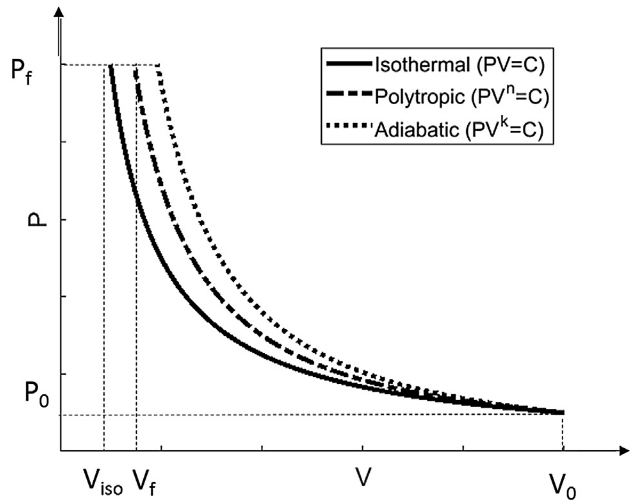

Fig. 3. Pressure-volume plot indicating isothermal, polytropic, and adiabatic compression.

air volume as 12 cm.

3.2. Spray injection setup

The liquid piston setup is equipped with an additional flow circuit for injecting water spray inside the compression chamber. A schematic of the spray injection setup is shown in Fig. 5. The system consists of pipes, a water pump, a pressure regulator, a flow meter, a spray nozzle, a power source, and a DAQ system. A positive displacement pump set to a maximum pressure of 100 psi (689 kPa) is used to drive the water spray into the chamber. The spray water loop is closed by the water pump with its inlet connected to the bottom of the outer cylinder and its outlet connected to the spray nozzle in the compression chamber. This ensures that the volume of air at the start of each compression cycle remains the same, and no secondary drain is needed for the additional water from the spray. When the pump is switched on, the spray injection starts in the chamber, and its injection pressure is controlled using a pressure regulator. To measure the flow rate of water through the spray loop, a flowmeter is connected after the pressure regulator. The flowmeter and pump are powered using a 12 VDC power supply, with the flowmeter readings being recorded by the DAQ. The spray nozzle is mounted on the top surface of the compression chamber and is radially centered to generate a symmetric spray pattern inside the chamber.

The design of the nozzle along with a spray angle affects the droplet dynamics and the resulting heat transfer characteristics. The spray nozzles can be selected based on the type of nozzle (spray pattern), droplet distribution, spray angle, injection pressure range, and the pipe size. As the compression chamber has a small air volume, a low flow-rate nozzle was selected, with a medium to coarse droplet distribution. Therefore, low-flow whirl nozzles of various spray angles (60°, 90°, and 120°) with a 0.043 in. (1.09 mm) orifice diameter from BETE [25] (Nozzle number WL ¼) were used. A schematic of the spray pattern for the selected type of nozzle is shown in Fig. 6. A full cone nozzle shown here was selected as it covers the maximum air volume during the compression process. The 1/8 in. pipe was used for the water line connecting to the spray nozzle. The injection pressure of spray was varied from 30 psi (207 kPa) to 70 psi (483 kPa) using the pressure regulator.

3.3. Measurement devices

Air pressure is measured using a pressure transducer (OMEGA PX309 −100 psi) mounted on the top plate of the outside cylinder along with the solenoid valves. The pressure transducer is a stainless-steel transducer that uses a high accuracy silicon sensor. Two thermocouples are placed inside the compression chamber to measure temperature. K-type 40-gauge diameter thermocouples are used which are placed in radially opposite directions about 1 in. from the center of the spray nozzle and 1/2 in. below the top surface of the compression chamber. To prevent direct injection of spray on to the thermocouples and affect the temperature readings, the thermocouples were protected by a thin cylindrical plastic shell. This ensured that the droplets did not directly interact with the thermocouples, however, indirect collisions due to splashing and rebounding water droplets could not be avoided.

For the spray cooling measurements, a pressure regulator (NORGREN R91W-2AK-NLN) controlled the injection pressure of spray, and the flowmeter used is OMEGA flr-1012. The measurement instruments and their corresponding accuracies are listed in Table 1. Before starting the experiments, the measurement devices were calibrated repeatedly to maintain their measurement accuracy.

These measurement instruments are connected to a DAQ system to measure various parameters during the compression and intake strokes. The DAQ system converts the analog voltage signals from the transducer and thermocouples into digital signals. A controller is used to automate the compression process for multiple cycles. The

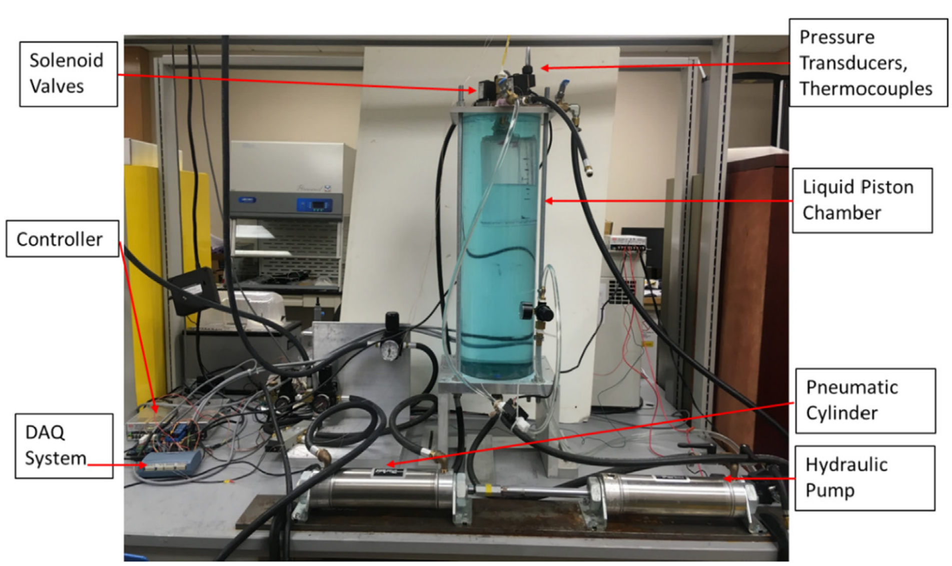

Fig. 4. A table-top setup of Liquid piston compressor.

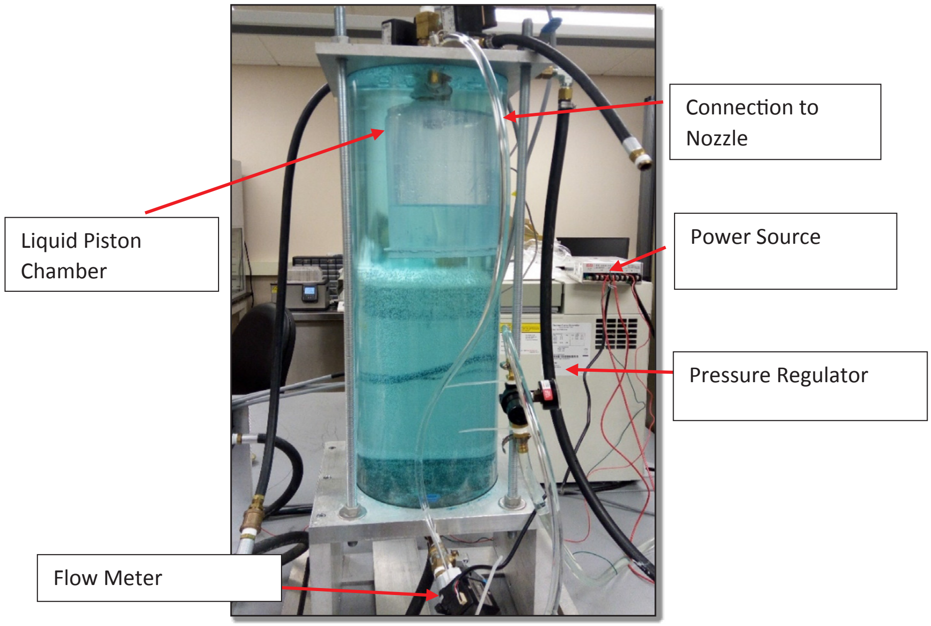

Fig. 5. Spray Injection Setup.

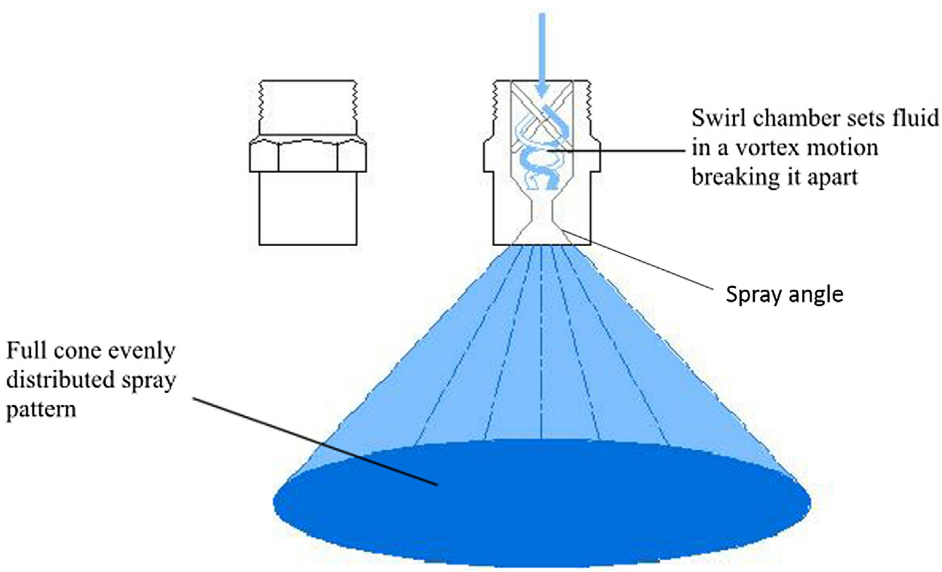

Fig. 6. Spray pattern with full cone whirl nozzle.

Table 1
Measurement devices and corresponding accuracies.

Measurement	Description	Accuracy
Pressure	Electronic Pressure Transducer 0–100 psig (0–689 kPa)	± 0.25%
Temperature	K –Type Thermocouple	± 2.2 °C
Volume	Location Sensor	± 0.001 in.
Flowmeter	Electronic Flowmeter 0.5–5 L/min	± 3%

instantaneous air pressure in the chamber, the air temperature in the chamber, displacement of the piston in pneumatic cylinder, and the flow rate of the water in the spray circuit are recorded using a LabVIEW program. The location of piston in the pneumatic cylinder which is connected to the hydraulic pump is used to calculate the instantaneous air volume in the chamber by subtracting the amount of water added in the chamber from the initial air volume of the chamber.

3.4. Range of experiments

A parametric study is performed to investigate the effect of water spray in liquid piston compression process. The parameters studied in the experiments are spray nozzle angle, stroke time of compression, and spray water injection pressure. The spray nozzle angles of 60°, 90°, and 120° are considered. For each spray angle, the stroke time of compression is varied in three levels with stroke time about 3 s (fast), about 4 s (medium) and, around 5 s (slow). The stroke time setting was done manually by adjusting the inlet source pressure of the pneumatic cylinder which dictated the speed of the coupled hydraulic pump system. Also, injection of spray at different injection pressures affects the stroke time marginally. Therefore, the selected three levels of stroke times vary slightly from the set time based on the set inlet source pressure of the pneumatic cylinder and the injection pressure of spray. With each spray angle and compression stroke time, the injection pressure is varied between 10 and 70 psi (69–483 kPa). Initially, the injection pressure was varied only from 30 to 70 psi (207–483 kPa) for the 60° spray angle but it was observed that there was not a significant difference in the temperature and pressure profiles for this injection pressure range. Therefore, the injection pressure range was expanded to 10–70 psi (69–483 kPa) for the 90° and 120° spray angles. A compression stroke runs until about 780 mL of water is added into the compression chamber from the liquid piston.

4. Results and discussion

4.1. Effect of injection pressure

The injection pressure of spray is one of the major influential parameters affecting the spray characteristics such as the flow rate of spray and the droplet size distribution for a given nozzle. In this subsection, experimental results for the variation of injection pressure from 10 psi (69 kPa) to 70 psi (483 kPa) for a 90° spray angle nozzle are presented. The injection pressure significantly changes the flow rate of spray during compression as observed in Fig. 7. The flow rate of the fluid is determined by the nozzle diameter and effective pressure difference across the nozzle. A higher injection pressure provides a higher pressure difference which results in a higher flow rate for a fixed diameter nozzle. As the compression proceeds, the flow rate decreases for any given injection pressure. This decrease is due to the back-pressure created on the spray nozzle when the chamber air pressure increases during compression; effectively reducing the pressure differential between the nozzle and compression chamber.

The motion of liquid piston from the bottom of the chamber towards the top drives the compression of air. The volume of air inside the chamber reduces approximately linearly with time as seen in Fig. 8. Since the total amount of liquid added in the chamber is the same in all the cases, the compression process ends at the same final volume. The higher injection pressure of spray makes the compression process slightly faster. As the speed of the liquid piston is proportional to the pressure differential between the hydraulic pump and chamber, the slightly higher speed with higher injection pressure suggests a reduction of chamber pressure at the same time point. The pressure measurements during the compression process shown in Fig. 9 also confirm the marginal reduction of chamber pressure with higher spray injection pressure at any time point. The maximum pressure during the compression stroke is observed for the case without any spray injection. On spray injection during compression, the final pressure attained is significantly lower compared to that of no spray. As pressure and temperature are coupled for a gas in a closed chamber, a lower pressure of gas points towards reduced bulk gas temperature with the injection of water spray.

The temperature measurements inside the chamber during the compression process are shown in Fig. 10. It should be noted that these temperature measurements are local therefore their utility is limited to the relative assessment about the effectiveness of spray injection for temperature abatement. There is a significant rise in the air temperature with the base liquid piston (no spray) case as the heat transfer rate of the base liquid piston is not enough to dissipate the heat of compression to the surrounding. The water droplets from spray injection absorb a significant portion of the heat of compression which helps in temperature abatement throughout the compression process. The large surface area of water droplets along with the high specific heat of water are major contributing factors in the heat transfer enhancement. A higher spray injection pressure results in a significant reduction in air temperature especially towards the end of compression; however, the marginal gain in the temperature abatement reduces as the injection pressure increases. For injection pressure up to 30 psi (207 kPa), there is a significant abatement in air temperature at the end of compression, whereas, any further increase in injection pressure above 30 psi (207 kPa) results in only a minute marginal benefit for the temperature abatement.

The pressure-volume plot is a useful tool to understand the work of compression and identify the effectiveness of spray injection to attain a near-isothermal compression. The effect of injection pressure of spray on the pressure-volume plots is shown in Fig. 11. For comparison, the reference adiabatic and ideal isothermal pressure-volume curves are also plotted. It is observed that for compression without spray (base liquid piston), pressure-volume curve shifts marginally towards the isothermal curve while injection of spray shifts the pressure-volume curve considerably towards the isothermal conditions. A zoomed-in view shown in Fig. 11 of the pressure-volume plots shows a perceptible distinction between the various curves. Higher injection pressures shift the curve towards near-isothermal conditions; however, the difference

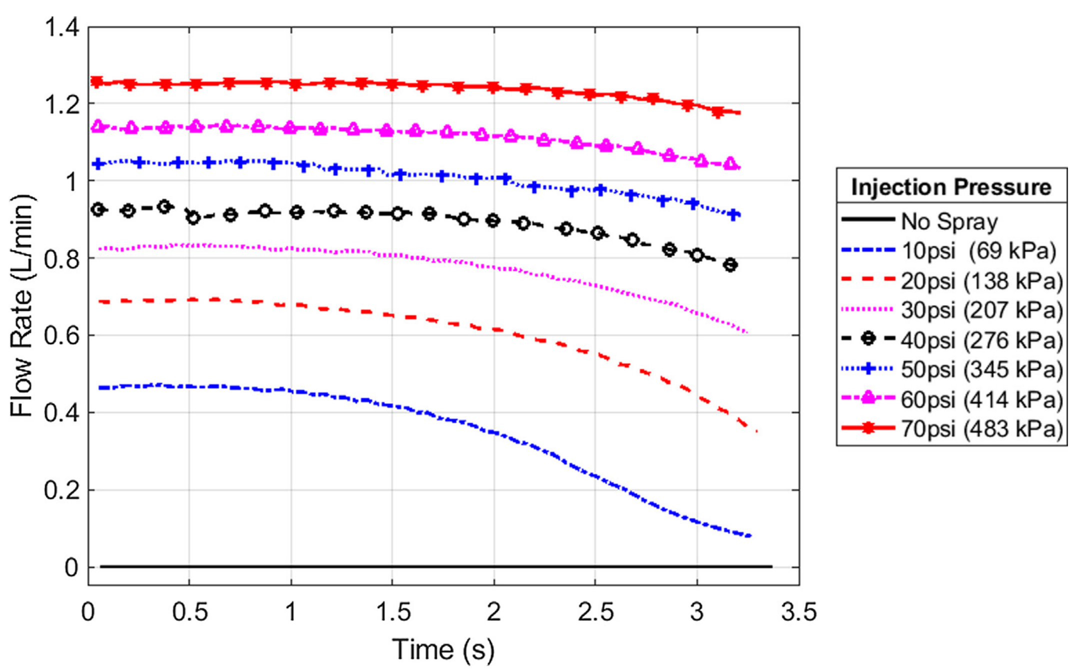

Fig. 7. Flow rates of the spray with different injection pressures during a compression cycle. (With 90° spray nozzle and fast level of stroke time).

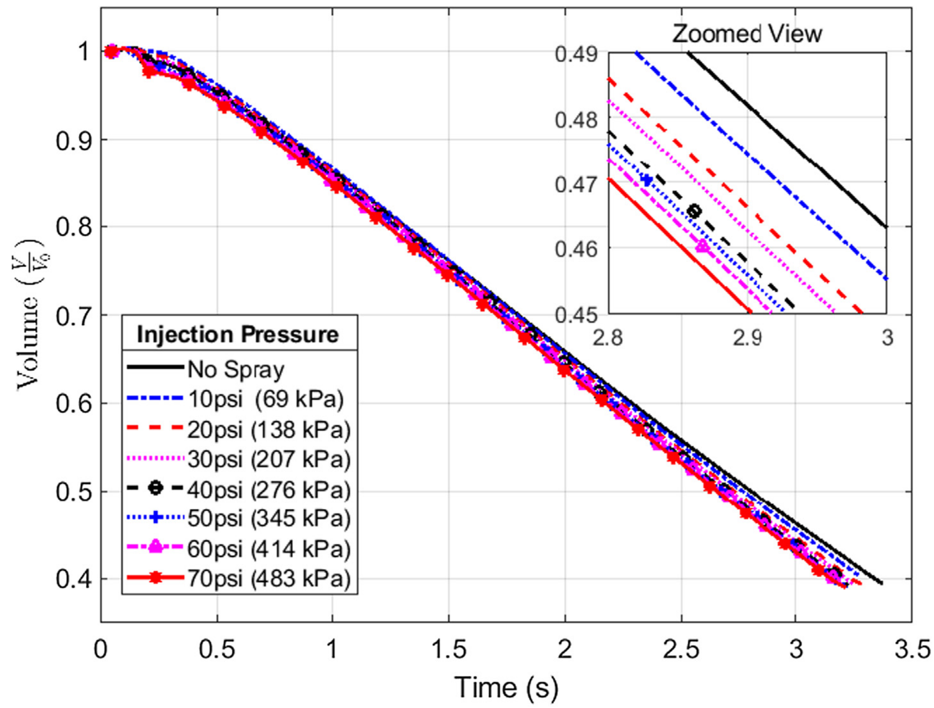

Fig. 8. Volume of air inside the chamber during compression relative to initial volume $V_0 = 1.28$ L in experiments of different spray injection pressures. (With 90° spray nozzle and fast level of stroke time).

in the curves is not significant for injection pressures of 40 psi (276 kPa) and higher. This indicates that an injection pressure of 40 psi (276 kPa) is good enough to absorb the bulk of the heat of compression and any further increase in injection pressure has a marginal benefit.

The effect of spray injection pressure during compression can also be analyzed through the compression work, calculated as an area under the pressure-volume curve. As spray injection shifts the pressure-volume curve towards an isothermal condition, the compression work reduces significantly compared to a compression process without spray. A higher spray injection pressure leads to a greater reduction in compression work, however, this comes with an additional cost of pump work to generate the spray at higher pressures. The compression work and spray work at different injection pressures of spray are summarized in Table 2. They are normalized by the product of the initial pressure and volume (P_0V_0). The normalized compression work reduces with an increase in spray injection pressure. The percentage reduction in

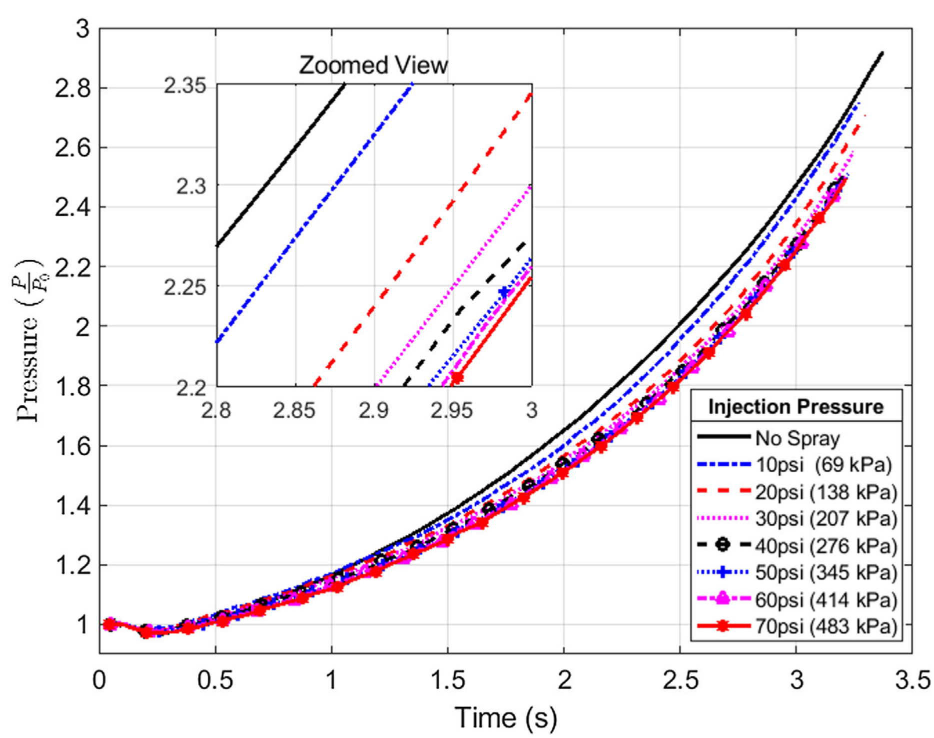

Fig. 9. Pressure inside the chamber during compression for different spray injection pressures. (90° spray nozzle and fast level of stroke time).

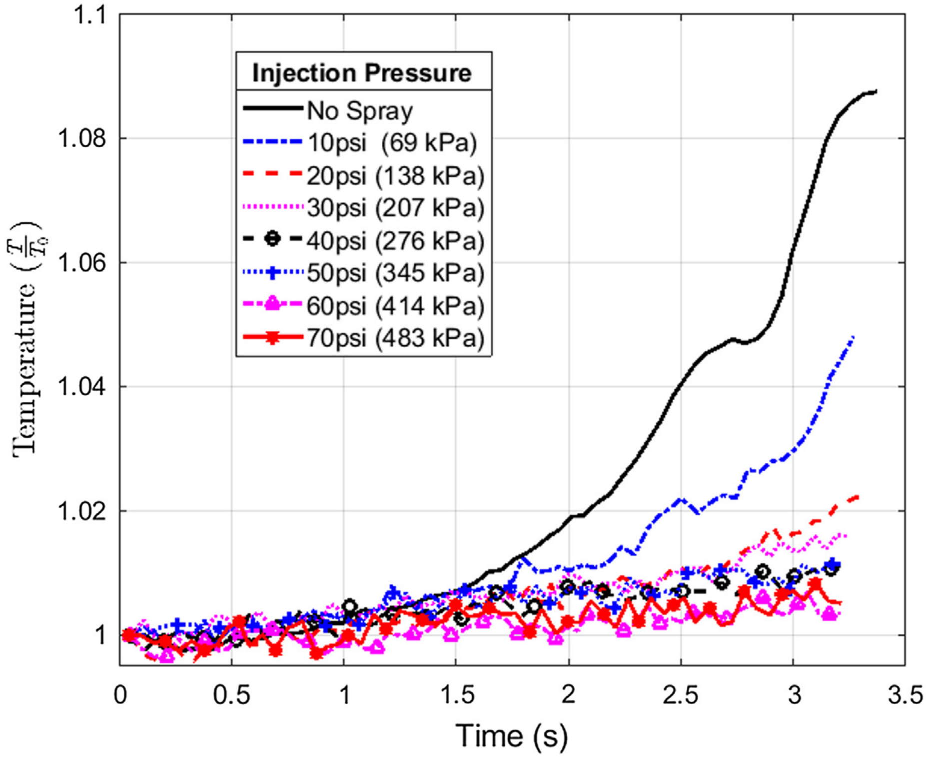

Fig. 10. Temperature plots during compression for different spray injection pressures. (With 90° spray nozzle and fast level of stroke time). *Note.* Temperature measurements are local point measurements not bulk gas temperature.

compression work with spray injection from the base (no spray) compression case is presented in column (4) of Table 2 to evaluate the degree of effectiveness of spray injection. Spray injection at 10 psi (69 kPa) reduces the compression work by about 10%, and a reduction of up to 25% can be achieved at higher spray injection pressures. However, this reduction is accompanied by a higher spray work as shown in columns (5) and (6) of Table 2. At lower injection pressures of spray, the spray work is a small fraction of the compression work, whereas, additional spray work as high as 60% of the compression work is needed at higher injection pressures. A comparison of column (4) and column (6) guides whether the additional spray work is justifiable to achieve a reduction in the network input. At injection pressures of up to

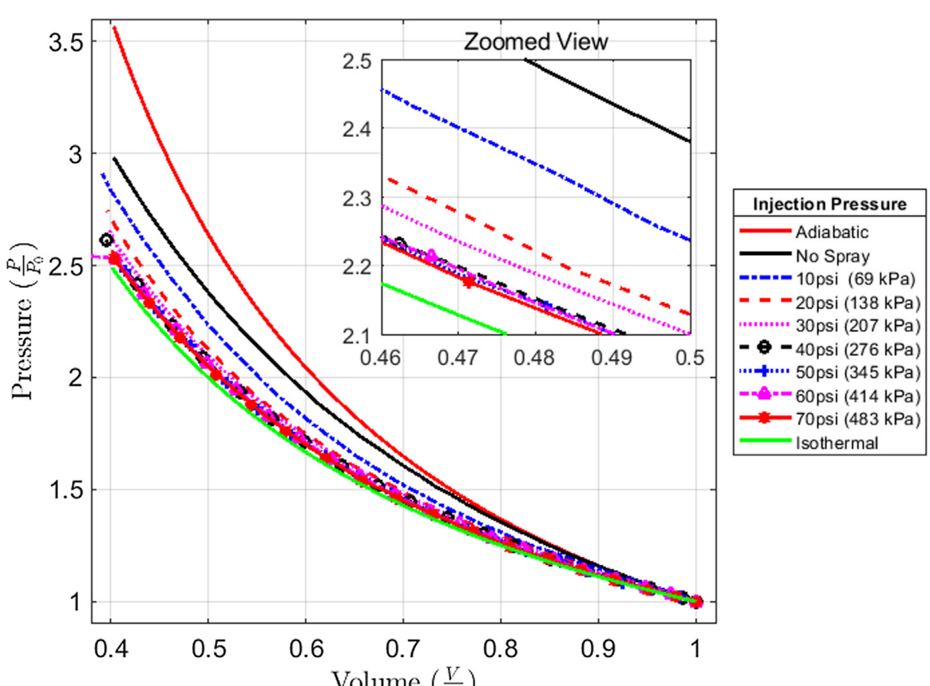

Fig. 11. Pressure volume plots during compression for different spray injection pressures in comparison with adiabatic and isothermal plots. (With 90° spray nozzle and fast level of stroke time).

Table 2
Compression work and Spray Work for compression at different injection pressures.

Injection Pressure of Spray (1)	Normalized[a] Compression Work[b] (2)	Reduction in Compression work with spray from No spray (3)	Percentage Reduction in Compression work from No Spray case (4)	Normalized[a] Spray Work[b] (5)	Spray-Work in percentage of No Spray compression work (6)
No Spray	0.4102	–	–	0	0
10 psi (69 kPa)	0.3683	0.0419	10.21%	0.0100	2.44%
20 psi (138 kPa)	0.3352	0.0749	18.27%	0.0340	8.28%
30 psi (207 kPa)	0.3230	0.0872	21.26%	0.0655	15.97%
40 psi (276 kPa)	0.3211	0.0890	21.71%	0.1008	24.58%
50 psi (345 kPa)	0.3139	0.0963	23.48%	0.1418	34.57%
60 psi (414 kPa)	0.3147	0.0954	23.27%	0.1878	45.79%
70 psi (483 kPa)	0.3093	0.1009	24.60%	0.2441	59.51%

Notes.
[a] Normalized with P_0V_0.
[b] Average of ten cycles and the standard error is within 2%.

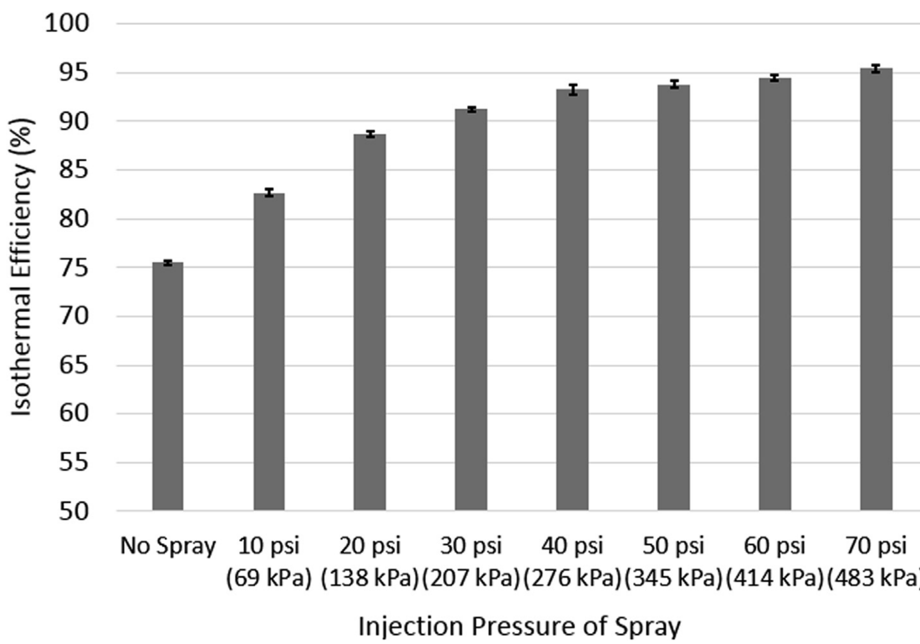

Fig. 12. Isothermal efficiency of compression with spray of different injection pressures. (With 90° spray nozzle and fast level of stroke time).

30 psi (207 kPa), the reduction in compression work achieved is greater than the additional spray work required. However, at injection pressures of 40 psi (276 kPa) and above, additional spray work exceeds the reduction in compression work. This indicates that spray up to 30 psi injection pressures are worthwhile to achieve a net reduction in work when the spray injection is used with its own operating pump. Furthermore, it can be observed that the highest gain (difference in column 4 and column 6 of Table 2) is achieved with 20 psi injection pressure. Therefore, the maximum net reduction in work is achieved when the injection pressure is lower than 30 psi (207 kPa).

Potentially, the hydraulic pump can be replaced with a spray injection system to supply the compression work. In such a system, the motion of liquid piston during the compression stroke would be achieved with the injection of water from the spray. During the air intake stroke of compression, the liquid needs to be removed from the compression chamber to bring the liquid level to the initial stage. This can be achieved by operating two liquid piston compression systems simultaneously with a phase difference, i.e. during the suction of one liquid piston, compression is performed in another system. In this way, the spray injection system can be effectively used to achieve the motion of the liquid piston in both the intake and compression stroke of the compression process. From Table 2, it can be observed that spray injection can provide the necessary compression work since the normalized spray work is of the same order as that of the normalized compression work. Further studies into the design of a spray system to match the spray work with the compression work are needed. Such a system would be highly effective because the injection of spray would not account for much additional work requirements.

Isothermal efficiency of compression assesses the effectiveness of a compressor system to achieve a near-isothermal compression. The polytropic index of compression is evaluated by fitting a polytropic process model through the pressure-volume plot to estimate the isothermal efficiency. Fig. 12 shows the isothermal efficiency of compression at various spray injection pressures. The error bar for each of the cases represents the standard deviation in the efficiency values for ten compression cycles. The base liquid piston without spray injection shows an isothermal efficiency of about 75%. The efficiency increases consistently with a higher spray injection pressure and approaches close to 95% efficiency. When the compression process shifts towards an isothermal condition, a reduction in the compression work and cooling work is observed. This results in a significant improvement in the isothermal efficiency. However, the gain in efficiency due to additional injection pressure reduces consistently indicating diminishing efficiency gain with higher injection pressure. Furthermore, the small error bars indicate that a consistent improvement in efficiency is possible with spray injection. Overall, spray injection is effective in achieving an

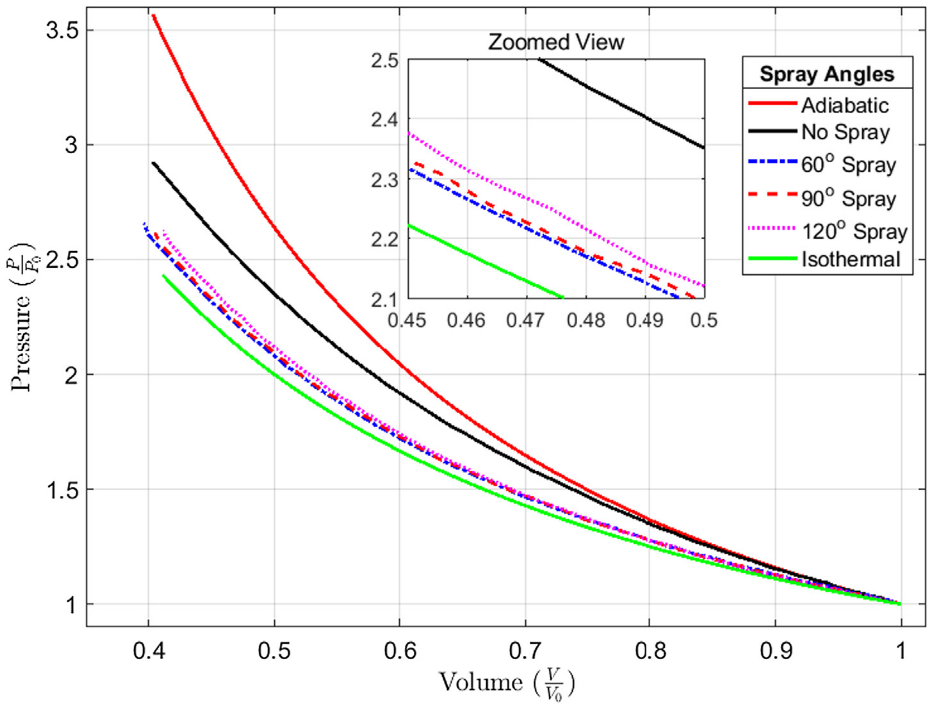

Fig. 13. Pressure-volume plots with spray injection of different spray angles during compression. (With 30 psi injection pressure and medium level of stroke time).

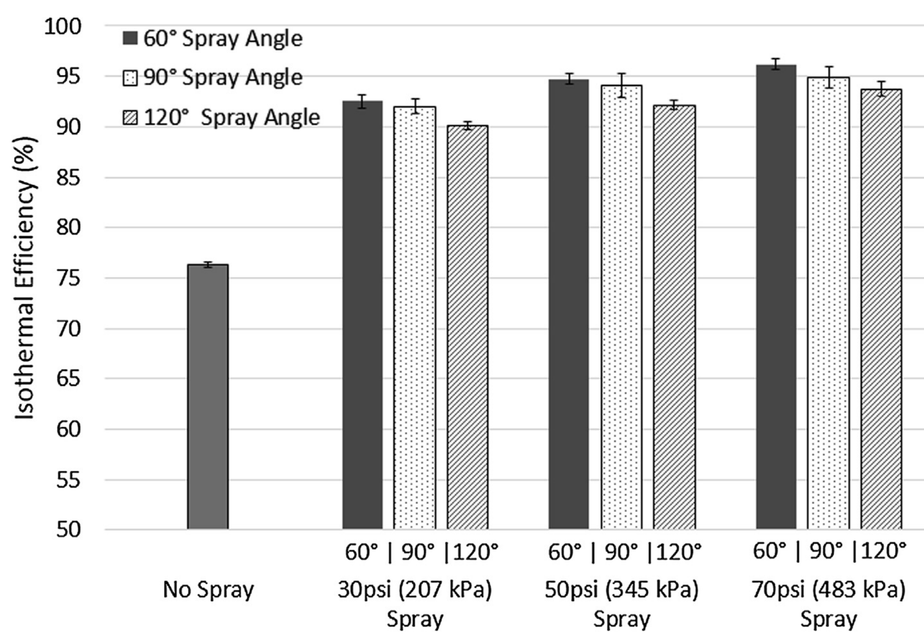

Fig. 14. Isothermal efficiency with spray injection from different nozzle angles for various injection pressures. (With medium level of stroke time).

isothermal efficiency of up to 95% in a liquid piston compressor.

4.2. Effect of spray nozzle angle

The droplet characteristics during spray generation are heavily influenced by the injection pressure and the nozzle angle. From the manufacturer's data[1] on the droplet size distribution of spray, it is noted that increasing the spray angle for a constant injection pressure decreased the mean droplet diameter. The spray angle also determines the spread of the droplets in the chamber volume. A smaller spray angle injects the majority of droplets in the middle of the chamber whereas a wider angle injects a large amount of spray on the walls of the chamber. In the current set of experiments, a higher fraction of spray droplets will hit the chamber walls with a 120° spray angle than the 90° and 60° spray angles which may affect heat transfer characteristics of spray. The results are presented for the experiments done with spray angles of 60°, 90°, and 120° for a compression stroke time of about 4 s (medium stroke).

Pressure-volume plots in Fig. 13 for different spray angles indicate that all three spray angles are equally effective in shifting curves towards an isothermal condition. Although the spray angles are quite different, the flow rates of spray are of a comparable magnitude for

[1] Note: Spray droplet distribution characteristics can be requested from BETE Fog Nozzle, Inc. [25] for nozzle number WL 1/4.

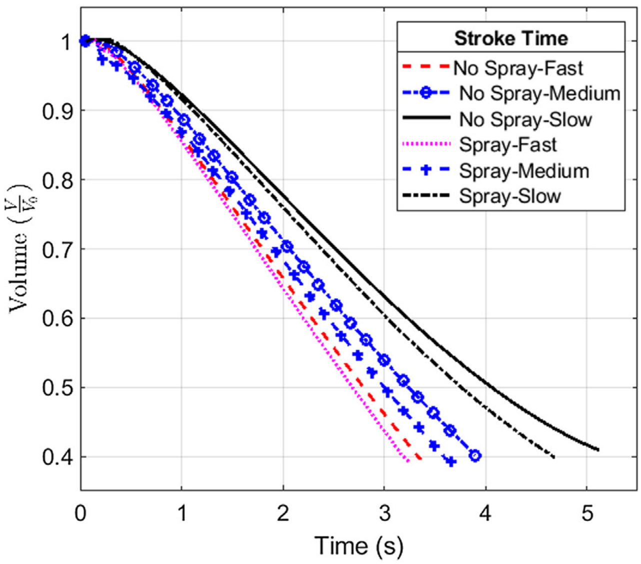

Fig. 15. Volume of air in the chamber during compression process relative to initial volume V_0 = 1.28 L for different stroke times. (With 30 psi injection pressure and 90° spray angle).

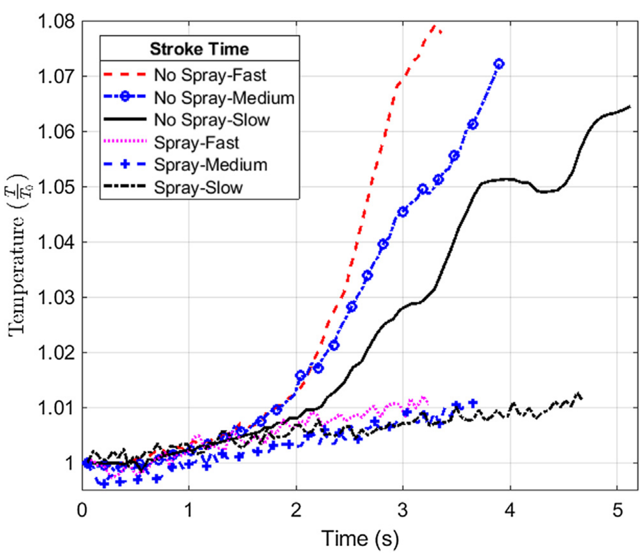

Fig. 16. Temperature plots during compression with and without spray injection for different stroke times. (With 30 psi injection pressure and 90° nozzle angle). *Note.* Temperature measurements are local point measurements not bulk gas temperature.

these cases due to the same injection pressure. Theoretically, the spray from the 120° nozzle angle should have a higher heat transfer area due to the overall smaller droplets compared to 90° and 60°. However, a slight difference in the pressure-volume curves of different spray angles suggests that the nozzle angle has only a marginal effect on the heat transfer characteristics of the spray. The curves for 60° and 90° spray angles appear slightly closer to the isothermal condition than the curve for the 120° angle suggesting that the 60°- 90° angle range is favorable for a lower compression work in the given chamber.

The isothermal efficiency of compression for all three spray angles (60°, 90°, and 120°) at injection pressures of 30 psi (207 kPa), 50 psi (345 kPa), and 70 psi (483 kPa) for the medium compression stroke are presented in Fig. 14. The error bar for each of the cases represents the standard deviation of ten observations. For any injection pressure, the variation in the spray angle marginally affects the isothermal efficiency. Also, a spray angle of 60° is slightly better than a spray angle of 90°, and 120° is the least efficient among all the three angles for any injection pressure. With a larger nozzle angle, a higher percentage of droplets impact the walls of the chamber compared to the smaller angles effectively reducing droplets in contact with the air. This results in lower heat transfer rate for larger spray angles consistently across all injection pressures, therefore, the isothermal efficiency decreases with an increase in the nozzle angle at all injection pressures. Even though the droplets become finer with the larger spray angle, an overall reduction in isothermal efficiency suggests that an optimal spray angle which can generate finer droplets with a minimum loss of droplets is desired for maximum efficiency. Noticeably, in these experiments for different nozzle angles, a 60° nozzle shows around a 1–4% improvement in the isothermal efficiency compared to a 120° nozzle irrespective of the spray injection pressure.

4.3. Effect of stroke time of compression

An efficient compressor system with a high-power density is highly desirable for various compressor applications. High-power density compression requires a small stroke time of compression. However, the heat transfer characteristics of liquid piston compressor alter with the change in the stroke time of compression. A faster compression (smaller stroke time) allows less time for effective heat transfer compared to a slower compression process (larger stroke time). Thus, a slower compression is favorable to realize a near-isothermal compression, though a slow compression results in a low power density. An ideal case would be a near isothermal compression process along with a high-power density system. The ability of spray injection to achieve near-isothermal compression with a fast compression stroke is investigated in this section. Spray injection at various levels of stroke time of compression is examined in three stages (slow, medium, and fast compression). The spray injection at 30 psi (207 kPa) injection pressure using a 90° spray angle is presented in this section.

The volume of air in the chamber during the compression process of different stroke times shows an approximately linear trajectory in all the cases as shown in Fig. 15. As expected, it takes a longer time to reach the final volume with the slower compression stroke and the spray injection cases show slightly faster compression compared to the corresponding no spray case. The air temperature during the compression of different stroke times is shown in Fig. 16. For compression in the liquid piston without spray injection, the peak air temperature increases with a shift towards faster compression. A shorter stroke time allows less time for heat dissipation which results in a comparatively higher air temperature than with the longer stroke times. The fact that the temperature profiles for all spray injection cases are very similar indicates that spray injection is equally effective for temperature abatement irrespective of the stroke time. Spray injection absorbs a significant portion of the heat of compression and lowers the air temperature significantly. With spray injection, a slower stroke time does not provide significant improvements in heat dissipation as the water droplets absorb the majority of the heat. Therefore, even with a faster compression process, spray injection is equally effective in maintaining a near-isothermal condition.

Further investigations at different stroke times of compression are performed using pressure-volume plots. For compression without spray injection, the pressure-volume plots in Fig. 17 indicate that a slower compression shifts the curve slightly towards the isothermal curve. Therefore, the compression work reduces with an increase in stroke time. However, as a slower stroke time hampers the power density, compression with a faster stroke time is desired for a high-power density. Compression with spray injection balances both characteristics. The pressure-volume plots with spray injection are closer to one another and tend towards the isothermal conditions. This implies that a significant reduction in compression work is possible with spray injection irrespective of the stroke time. A zoomed-in view in Fig. 17 shows that spray injection with slow compression is marginally closer

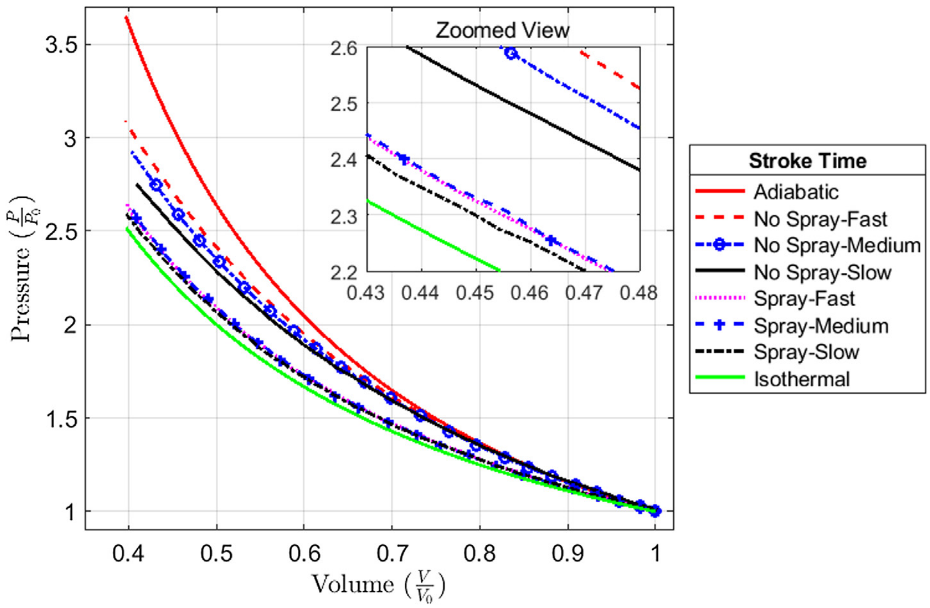

Fig. 17. Pressure-volume plots with and without spray injection for different stroke times. (With 30 psi injection pressure and 90° nozzle angle).

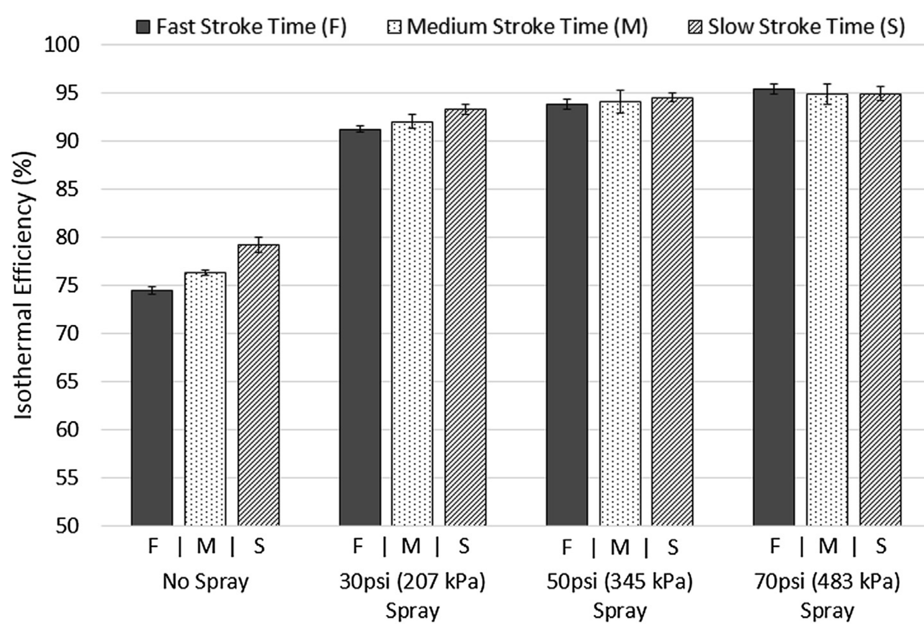

Fig. 18. Isothermal efficiency with and without spray injection at various stroke time of compression for various injection pressures. (With 90° nozzle angle).

to the isothermal curve, however, the difference is indistinguishable. This indicates that a faster compression process can be achieved with its compression work comparable to that of a slower compression process.

The isothermal efficiency with spray injection at different compression stroke times is evaluated for various injection pressures. The isothermal efficiencies for the three levels of compression stroke time (fast, medium, and slow compression) at different injection pressures of 30 psi (207 kPa), 50 psi (345 kPa), and 70 psi (483 kPa) are presented in Fig. 18. The error bar for each of the cases represents the standard deviation of efficiencies evaluated for ten compression cycles. A base liquid piston case without spray injection displays an isothermal efficiency of around 75% with a faster compression stroke. The efficiency of the base liquid piston can be improved by up to 80% with a slower compression process. However, as mentioned earlier, this affects the power density of the compressor and is therefore not worthwhile. Spray injection improves the efficiency significantly for the varied stroke times at given injection pressures. At a lower injection pressure (207 kPa), slowing the compression improves the efficiency marginally. However, at a higher injection pressure (483 kPa), an isothermal efficiency of about 95% is observed irrespective of the level of stroke time. At a higher injection pressure, a faster compression is highly effective in achieving an efficient compression process with a high power density.

Although this study is limited to low compression ratios (about 2.5), spray injection could be effectively extended to larger compression ratios. The numerical study [20] of droplet heat transfer in liquid piston suggests higher than 95% compression efficiency for a compression ratio of 10. However, the study also highlights a decrease in efficiency at a higher compression ratio. The high air density in the chamber at larger compression ratios makes it harder for the spray to penetrate a sufficient distance or diffuse spray droplets throughout the chamber volume. This necessitates a significantly higher spray injection pressure for high compression ratios. Additionally, the heat of compression would be significantly larger at higher compression ratios, therefore, a higher mass loading of spray would be required to achieve a high heat transfer rate. These additional requirements would certainly require extra work input from the spray pumps. Techniques like trajectory optimization [26], multi-stage compression, and multi-nozzle configuration [27] could be valuable to use spray efficiently at higher compression ratios. An experimental investigation [23] of water injection (50 bar injection pressure) in conventional piston reciprocating compressor for a compression ratio of 25 has shown promising results to keep air temperature below 100 °C with 28% work saving relative to adiabatic compression. However, achieving isothermal efficiency higher than 95% for higher compression ratios is challenging, and continued investigations are needed to make it a reality.

5. Conclusions

Water spray injection is studied experimentally in a liquid piston compressor for a pressure ratio of about 2.5. Temperature, pressure, and volume of air are measured during the compression process to assess the effectiveness of spray injection. The study was designed to include a range of experiments covering various injection pressures of spray, different spray angles, and various compression stroke times. For each experimental case, at least ten continuous compression cycles were performed. Measurements showed cyclic consistency in pressure and temperature data during the compression process with and without spray injection.

The injection of spray is observed to be very effective in reducing the air temperature during the compression process. The temperature abatement increases with an increase in injection pressure of spray from 10 psi (69 kPa) up to 70 psi (483 kPa). With the increase of injection pressure, the flow rate of spray water increases which provides a higher rate of heat transfer and helps reduce air temperature significantly. The pressure–volume plots close to isothermal curves are observed with the spray above 30 psi (207 kPa) injection pressures. The spray injection shows up to a 25% reduction in compression work from the base liquid piston compression, however, the spray work also increases with injection pressure. Moreover, the isothermal efficiency of compression can be improved significantly with the injection of spray. The improvement in efficiency increases with the increase of spray injection pressure. The injection of spray at 70 psi leads to 19–21% improvement in isothermal efficiency of compression from the base level efficiency of 75% for liquid piston without spray. Overall, water spray injection is highly effective in achieving near-isothermal compression with up to 95% isothermal efficiency.

Furthermore, experiments with different spray angles of nozzles were performed at 60°, 90°, and 120° nozzle angles for different injection pressures. It was observed that a 60° nozzle angle is marginally better than 90°, and 120° is least efficient at all the injection pressures of spray. The wider nozzle angles could lead to the loss of water droplets due to the impact of spray on the chamber wall and reduce the effective rate of heat transfer from the spray. An optimal spray angle specific to the chamber geometry that can create finer droplets with a minimum loss of droplets due to the impact on chamber wall should be considered for maximum efficiency.

Likewise, the usefulness of spray injection at various stroke times of compression was investigated by varying the speed of compression at three levels (fast, medium, and slow compression) at different spray injection pressures. It is observed that slower compression provides only marginal benefit in the efficiency at lower injection pressures, and isothermal efficiency of about 95% is observed irrespective of the speed of compression at a high injection pressure (70 psi). Therefore, spray injection is highly effective to attain efficient compression with a high-power density. Overall, water spray injection is a highly effective technique to achieve near-isothermal compression in the liquid piston. However, this study is based on experimentation at a relatively low compression pressure ratio, and further investigations are needed to confirm the effectiveness of spray injection at higher compression pressure ratios. Also, contamination of water vapor in the air could be an issue for many compressor applications. The effect of contamination of air on operation and maintenance of the compressor system needs to be further explored.

Declaration of Competing Interest

The authors declared that there is no conflict of interest.

Acknowledgment

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